MIL-E-37303B
24 August 1978
SUPERSEDING
MIL-E-37303A
28 May 1976

MILITARY SPECIFICATION

EPINEPHRINE HYDROCHLORIDE OPHTHALMIC SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Epinephrine Hydrochloride Ophthalmic Solution.

1.2 Classification. The solution covered by this specification shall be of the following strengths (see 6.2):

Strength 1 - 1 percent, 15 ml
Strength 2 - 2 percent, 15 ml

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for prosposals, form a part of this specication to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 202640-21T10:24Z
Check the source to verify that this is the currentversion before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material.

3.1.1 For strength 1. Shall be a sterile aqueous ophthalmic solution of levo-epinephrine base, 1.0 percent (plus 10 percent excess), as the hydrochloride salt, within the designated assay limits. The ophthalmic solution shall contain suitable antimicrobial agents and may contain stabilizers and buffering agents.

Source: https://assist.dla.mil -- Downloaded: 202640-21T10:24Z
Check the source to verify that this is the current version before use.

3.1.2 For strength 2. Shall be a sterile aqueous ophthalmic solution of levo-epinephrine base, 2 percent (plus 10 percent excess) as the hydrochloride salt, within the designated assay limits. The ophthalmic solution shall contain suitable antimicrobial agents and may contain stabilizers and buffering agents.

3.2 Finished product.

3.2.1 pH. The solution shall have an adjusted pH not to exceed 3.30 for strength 1 and 3.20 for strength 2 at time of manufacture and not less than 2.20 and not more than 3.50 until the end of the expiration dating period, when determined potentiometrically at 25°C, using the USP method.

3.2.2 Particulation. The solution shall meet the NF requirements for ophthalmic solutions for particulation in that the material shall be essentially free from foreign particles.

3.2.3 Assay.

3.2.3.1 l-Epinephrine base. The solution shall assay to contain not less than 90.0 percent and not more than 115.0 percent of the required amount of the l-Epinephrine base when determined by a suitable, accurate, and reproducible method.

3.2.3.1.1 The bidder or offeror shall submit, with its bid or proposal, the proposed method of assay for the finished ophthalmic solution. The method shall be subject to approval and acceptance by the Government. If the method is approved and accepted by the Government, such method shall be employed in the testing of the material.

3.2.4 Sterility. The ophthalmic solution shall be sterile when tested in accordance with the USP sterility test method.

3.2.5 Identification. Methods of identification for the active ingredient in the finished ophthalmic solution shall be submitted with the bid or proposal and shall be subject to approval and acceptance by the Government. If the method is approved and accepted by the Government, such method shall be used to identify the active ingredient. No further submission of the method is required, provided the contractor states that the product and identification tests have not been modified in any way. In addition, an identity test for the hydrochloride shall be performed to verify that the hydrochloride salt was used.

3.3 Ingredients. All ingredients used in the manufacture of the solution shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto, or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.4 Expiration dating period. The expiration dating period shall be not less than 30 months for strength 1 and not less than 21 months for strength 2.

Source: https://assist.dla.mil -- Downloaded: 202640-21T10:24Z
Check the source to verify that this is the currentversion before use.

3.5 Delivery. Not less than 27 months of the expiration dating period for strength 1, and not less than 18 months of the expiration dating period for strength 2 shall remain at the time of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of the inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Immediate container
Minor	II	2.5	Immediate container

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:24Z
Check the source to verify that this is the current version before use.

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
pH	3.2.1	S-2	5 composite
Assay	3.2.3	S-2	5 sample
Sterility	3.2.4	---	--
Identification	3.2.5	S-2	5
Ingredients	3.3	2/	--

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished ophthalmic solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Solution not essentially free from particulation or foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:24Z
Check the source to verify that this is the current version before use.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests. Tests shall be performed to determine compliance with all requirements of this document and all referenced documents.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 15 ml, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available drop-dispensing immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue of one strength only shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twelve intermediate packages (144 units) of one strength only shall be packed as specified in 5.4.4 of PPP-C-186.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:24Z
Check the source to verify that this is the current version before use.

5.3.2 Level B. Twelve intermediate packages (144 units) of one strength only shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commerical practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking shall be shown as specified in MIL-STD-129.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking shall be shown as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:24Z
Check the source to verify that this is the current version before use.

6.2 This specification covers the following items:

NATIONAL STOCK NO.	ITEM IDENTIFICATION	SPECIFICATION DESIGNATION
6505-00-080-4987	EPINEPHRINE HYDROCHLORIDE OPHTHALMIC SOLUTION, 1%, 15 ml	Strength 1
6505-00-935-9861	EPINEPHRINE HYDROCHLORIDE OPHTHALMIC SOLUTION, 2%, 15 ml	Strength 2

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1841

Source: https://assist.dla.mil -- Downloaded: 202610-21T10:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-E-37303B EPINEPHRINE HYDROCHLORIDE OPHTHALMIC SOLUTION

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-21T10:24Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

POSTAGE AND FEES PAID

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

DEFENSE PERSONNEL SUPPORT CENTER
DIRECTORATE OF MEDICAL MATERIEL, CODE-ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:24Z
Check the source to verify that this is the current version before use.